

ISA 634: Analytics Solution Deployment

Handout 03: Introduction to Linear Programming (Cont.)

Department of Information Systems & Analytics
Farmer School of Business
Miami University

Outline

- 1 Graphical Solution Method of an LP**
- 2 Graphical Visualization of Some LP Characteristics
- 3 Wyndor Glass Co. Problem via the Graphical Approach
- 4 Recap

Graphical Solution Method of an LP

Please take hand notes on the following topics:

- 1 Graphically draw and solve a 2-variable LP

Graphical Solution Method of an LP

Summary of the Graphical Solution Method

- **Draw the constraint boundary line for each constraint. Use the origin (or any point not on the line) to determine which side of the line is permitted by the constraint.**
- Find the feasible region by determining where all constraints are satisfied simultaneously.
- Determine the slope of one objective function line. All other objective function lines will have the same slope.
- Move a straight edge with this slope through the feasible region in the direction of improving values of the objective function. Stop at the last instant that the straight edge still passes through a point in the feasible region. This line given by the straight edge is the optimal objective function line.
- A feasible point on the optimal objective function line is an optimal solution.

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Graphical Visualization of Some LP Characteristics

Please take hand notes on the following topics:

- ① **Unbounded** feasible region and unbounded objective value
- ② **Infeasible** problem (i.e., no feasible solution exists)
- ③ **Uniqueness** of optimal solution
- ④ Summary
- ⑤ Key observations about **corner points & binding constraints**

Unbounded feasible region and unbounded objective value

Infeasible problem (i.e., no feasible solution exists)

Uniqueness of optimal solution

Uniqueness of optimal solution

Summary

Key observations about corner points & binding constraints

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 - Visualize Special Conditions in LP Models
- 4 Recap

Graphical Representation of Wyndor Glass Co. - Constraint #1

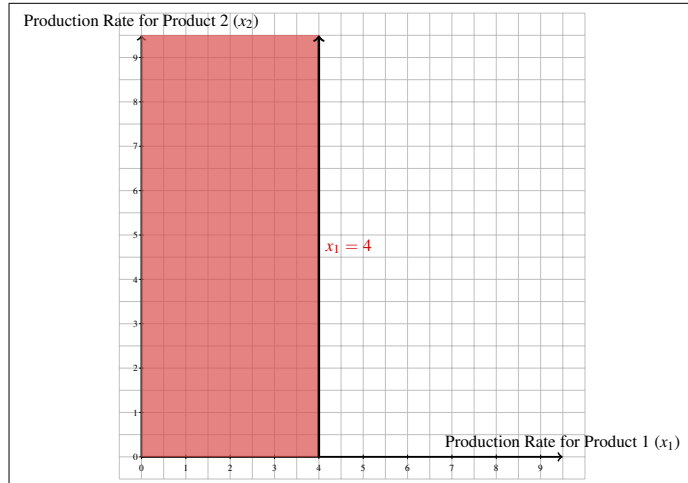


Figure: Nonnegative Solutions Permitted by $x_1 \leq 4$

Graphical Representation of Wyndor Glass Co. - Constraint #2

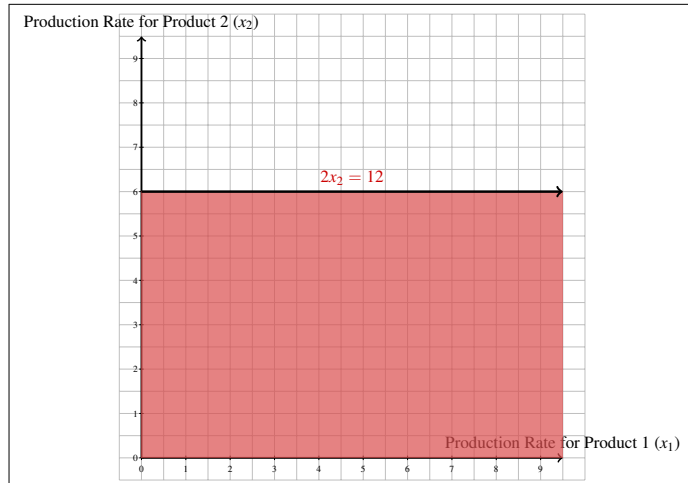


Figure: Nonnegative Solutions Permitted by $2x_2 \leq 12$

Graphical Representation of Wyndor Glass Co. - Constraint #3

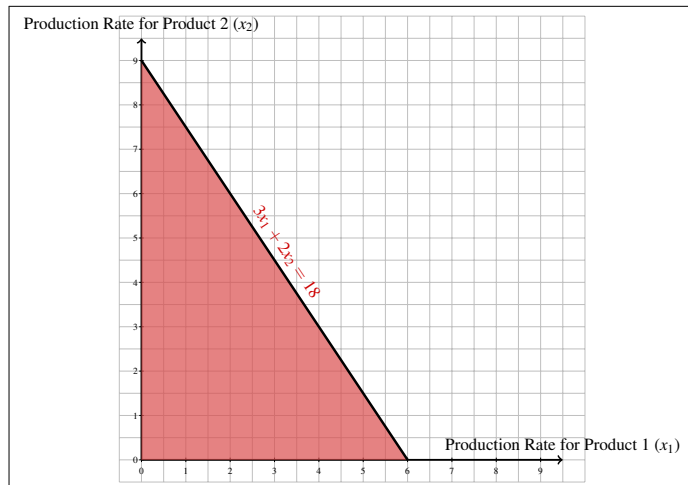


Figure: Nonnegative Solutions Permitted by $3x_1 + 2x_2 \leq 18$

Graphical Representation of Wyndor Glass Co. - Feasible Region

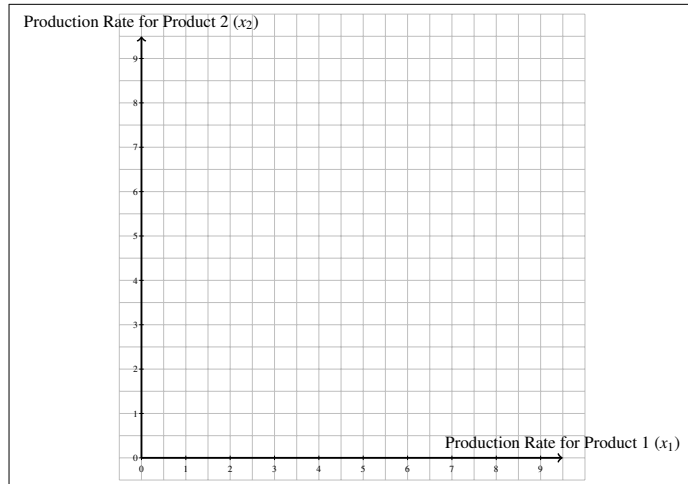


Figure: Nonnegative Solutions Permitted by Feasible Region

Graphical Representation of Wyndor Glass Co. - Feasible Region

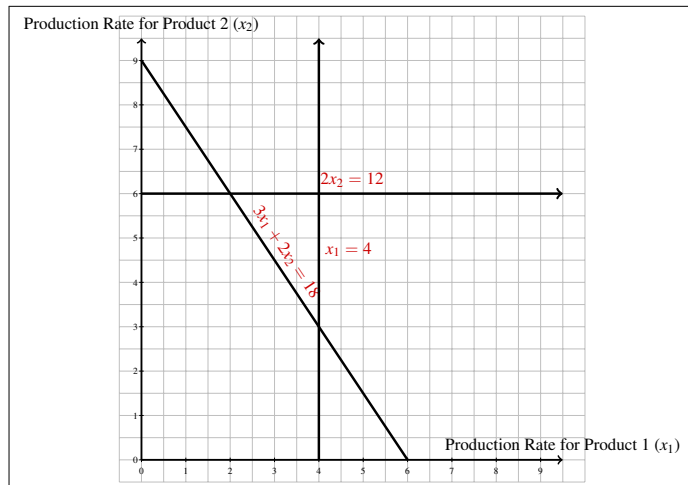


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Graphical Representation of Wyndor Glass Co. - Feasible Region

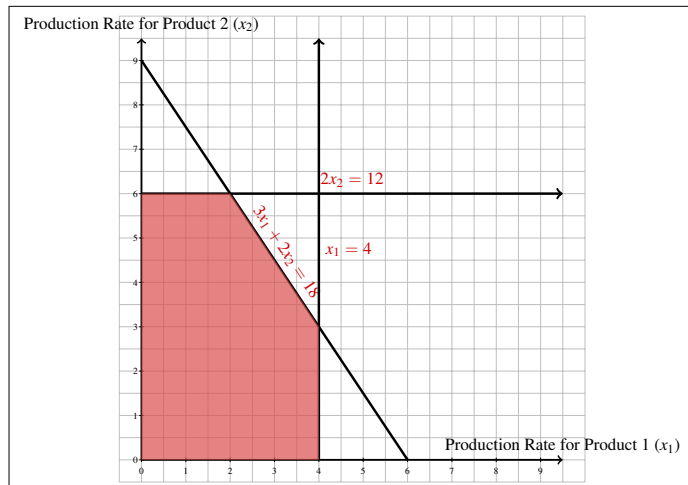


Figure: Nonnegative Solutions Permitted by Feasible Region

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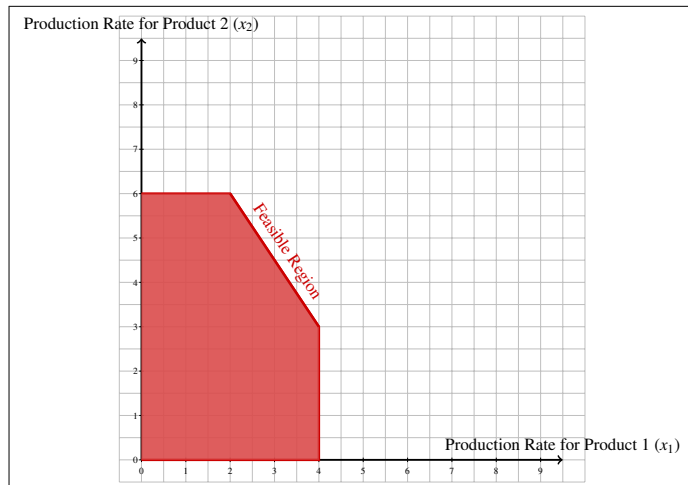


Figure: Nonnegative Solutions Permitted by Feasible Region

Graphical Representation of Wyndor Glass Co. - Find Optimal Sol.

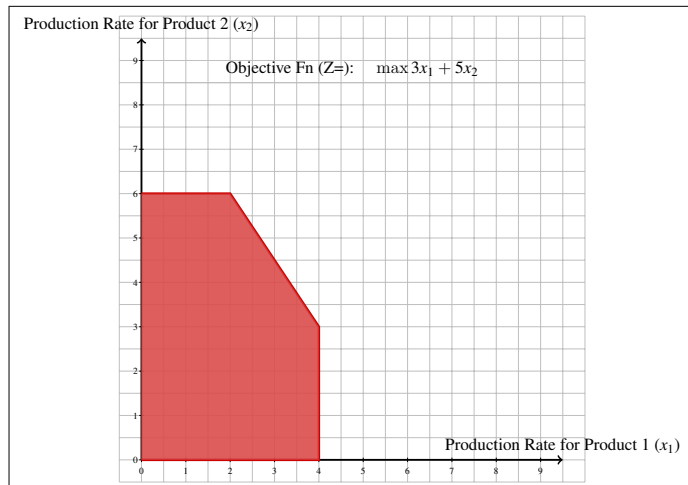


Figure: Finding the Optimal Solution

Graphical Representation of Wyndor Glass Co. - Find Optimal Sol.

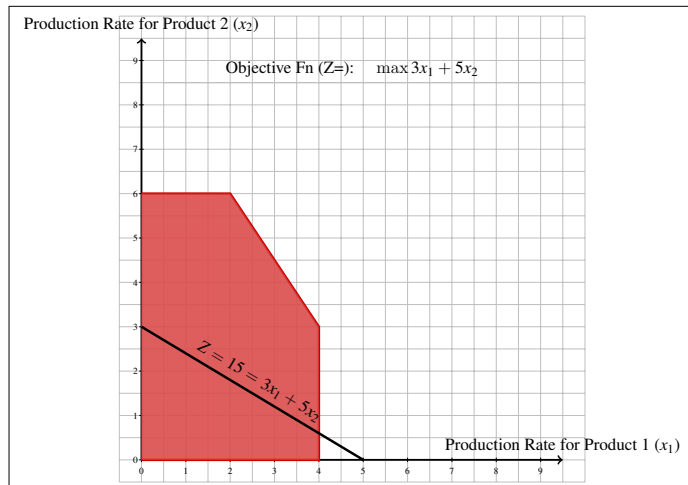


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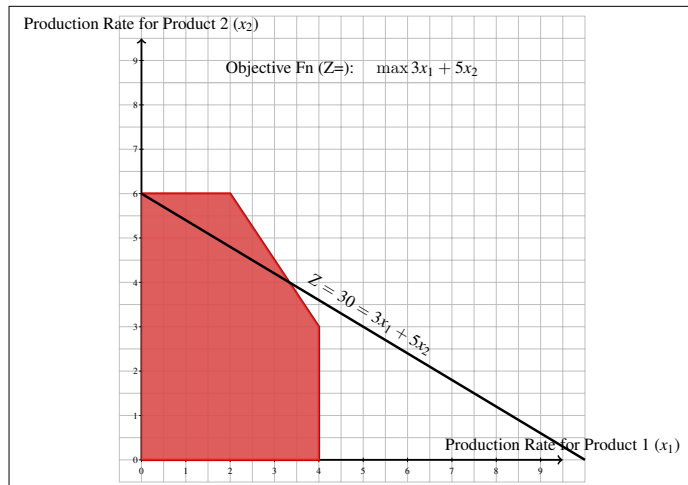


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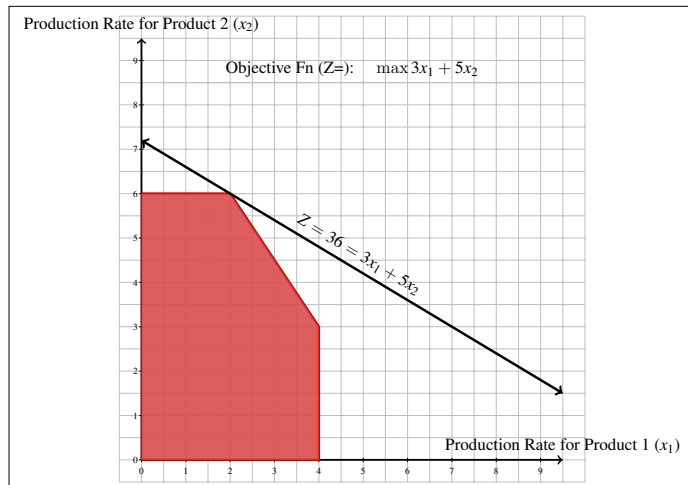


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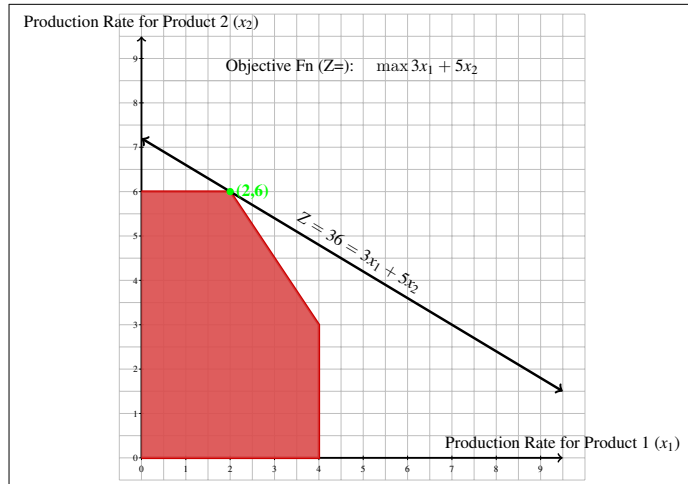


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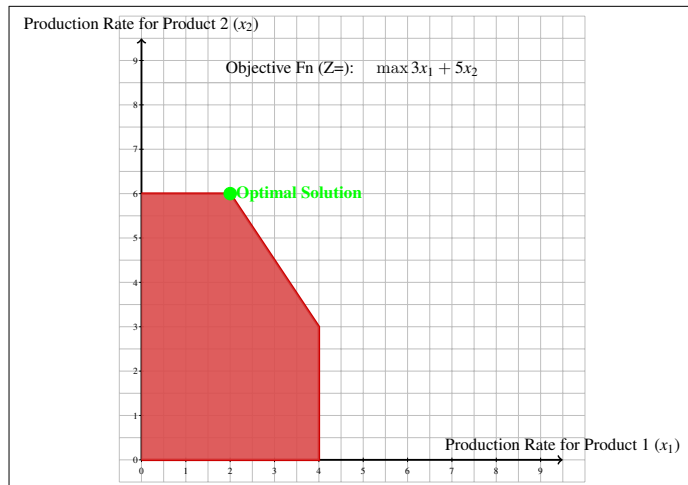


Figure: Finding the Optimal Solution

Questions based on the Wyndor Glass Problem

A) For the problem we have just discussed, are there multiple optimal solutions? Why/Why not?

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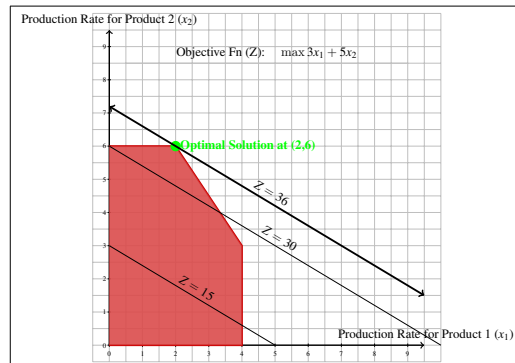


Figure: Parallel Lines Representing the slope-intercept form of Z , i.e., $x_2 = -\frac{3}{5}x_1 + \frac{1}{5}Z$.

Questions based on the Wyndor Glass Problem

B) What is the optimal solution if the profit per batch of Product 2 were changed to \$2,000? Is it a unique solution? Why/why not?

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Is it a unique solution? Why/why not?**

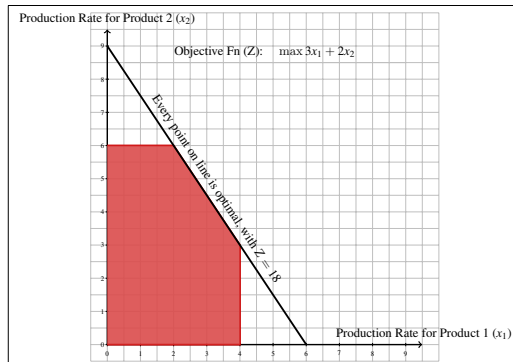


Figure: Multiple Optimal Solutions when profits for x_2 is changed to \$2,000

Questions based on the Wyndor Glass Problem

C) What happens to the feasible region (for the original problem) if we were to add the following constraint:

$$3x_1 + 5x_2 \geq 50$$

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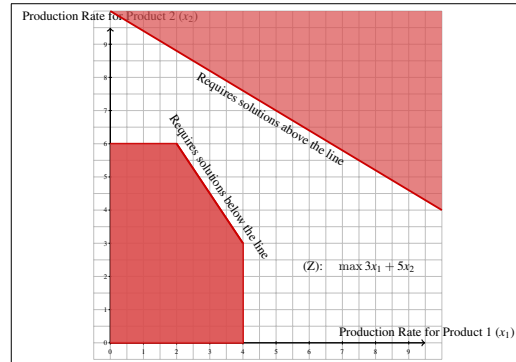


Figure: No Feasible Solutions

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A Summary of Main Learning Objectives In This Handout

Learning Objectives

- Solve LP problems (involving two decision variables) graphically
- Graphically visualize special conditions and observations of LP problems.